AARAV JAIN

Applied AI & Automation | Machine Learning | Agentic AI Systems
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EDUCATION

University of South Florida | Judy Genshaft Honors College | B.S. Computer Science | May 2027
GPA: 3.95/4.00 | Coursework: Machine Learning, Databases, Data Structures & Algorithms, Software Engineering

TECHNICAL SKILLS

Programming & Backend: Python, SQL, Java, C++, JavaScript, TypeScript, FastAPI, Flask, REST APIs, gRPC, Git, Docker
AI & ML: scikit-learn, pandas, NumPy, cross-validation, feature engineering, LLMs (Gemini, GPT-4, Llama 3.3), multi-agent systems, tool calling, RAG, embeddings, vector databases, YOLO
Data & Tools: ChromaDB, MongoDB, PostgreSQL, MySQL, data pipelines, model evaluation, Linux, Render

WORK EXPERIENCE

Cyber Florida at USF | Cybersecurity Analyst | Jan 2026 – Present
- Classify SOC alerts on suspicious traffic and malware C2 by correlating threat intelligence, recommending IP/domain blocks or user outreach
- Investigate simulated attack chains in Splunk, CrowdStrike and Windows logs, tracing attacker behavior and documenting incident findings
- Contributed to 2 threat intelligence advisories on the Vercel security incident and RDP exploitation with AI token theft

RARE Lab, USF | Undergraduate Researcher, AI & Robotics | Feb 2025 – Aug 2026
- Built a multimodal pipeline for Bloom, a companion robot: Vosk STT, Piper/OpenAI TTS, Gemini conversation logic, and real-time YOLO detection on a Hailo accelerator
- Developed bidirectional gRPC middleware between a robot and an LLM task planner, supplying live camera snapshots and structured JSON context for multi-step object manipulation
- Co-authored 3 robotics research papers on collaborative planning, robot abuse, and AR versus physical robots; 1 accepted and 2 submitted for review

HGrid | Software Engineer Intern | May 2025 – Aug 2025
- Automated hospital-patient agreement creation with PDF parsing and the DocuSign API, cutting a 3–4 hour manual workflow to ~20 seconds
- Integrated credit bureau data into score-based financing, interest-rate logic and agreement-template workflows
- Developed full-stack pagination for patient treatment records with configurable page sizes and bounded result sets

TECHNICAL PROJECTS

LLM Security Middleware | FastAPI, ChromaDB, Llama 3.3 | github.com/aadraxe/LLM-middleware-security
- Architected a three-layer detector combining regex rules, fuzzy heuristics, and Llama 3.3 70B semantic classification to screen jailbreak and prompt-injection attempts
- Benchmarked on 200 jailbreak prompts, reaching 94% precision and 65% recall versus under 10% recall for rules alone
- Built a RAG pipeline with embeddings and ChromaDB that scans retrieved chunks for indirect prompt injection

ContractMap | FastAPI, MongoDB, Gemini, scikit-learn | 3rd of 81 teams | ShellHacks 2026
- Owned the FastAPI backend, deployed on Render, including Gemini contract-PDF extraction and a ranking algorithm scoring 55 pairs across 182 projects by proximity, timing and savings
- Built a Gemini agent with function calling that searches projects, overlaps and shared resources, and proposes database writes that run only after user confirmation
- Trained random forest and ridge models on 205 and 44 filed projects with feature engineering, repeated cross-validation and error analysis; the cost model beat an industry estimate (27% vs 74% error)

Agentic Interview Helper | FastAPI, React, Llama 3.3 | github.com/jasminek12/agentic-ai-project
- Built most agents in a three-agent interview assistant (resume tailoring, adaptive interview practice, outreach) on FastAPI and Llama 3.3
- Designed shared conversation memory and stateful sessions so agents adapt follow-up questions and sessions can be restored
- Co-developed evaluation workflows scoring generated answers for relevance, clarity and depth against the job description

StoryBuddy | FastAPI, Groq, ElevenLabs, Web Speech | storybuddy-sagv.onrender.com
- Designed the backend speech-practice flow combining word recognition, pronunciation scoring, and ElevenLabs story narration
- Built retry logic allowing up to three attempts per practice word, advancing the story only after a correct spoken response
- Used Groq-hosted LLMs to generate progress reports on pronunciation accuracy, attempts and trends for parents and therapists